

PROJECT

5.2

VPC TRAFFIC FLOW AND SECURITY

NETWORKING SERIES

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Objective

This project enhances VPC security by setting up a Route Table, Security Group, and Network ACL. The route table directs traffic within the VPC, ensuring smooth data flow between subnets and the internet.

The security group acts as a firewall for instances, while the network ACL filters traffic at the subnet level. Together, they build a secure, scalable, and well-managed cloud network.

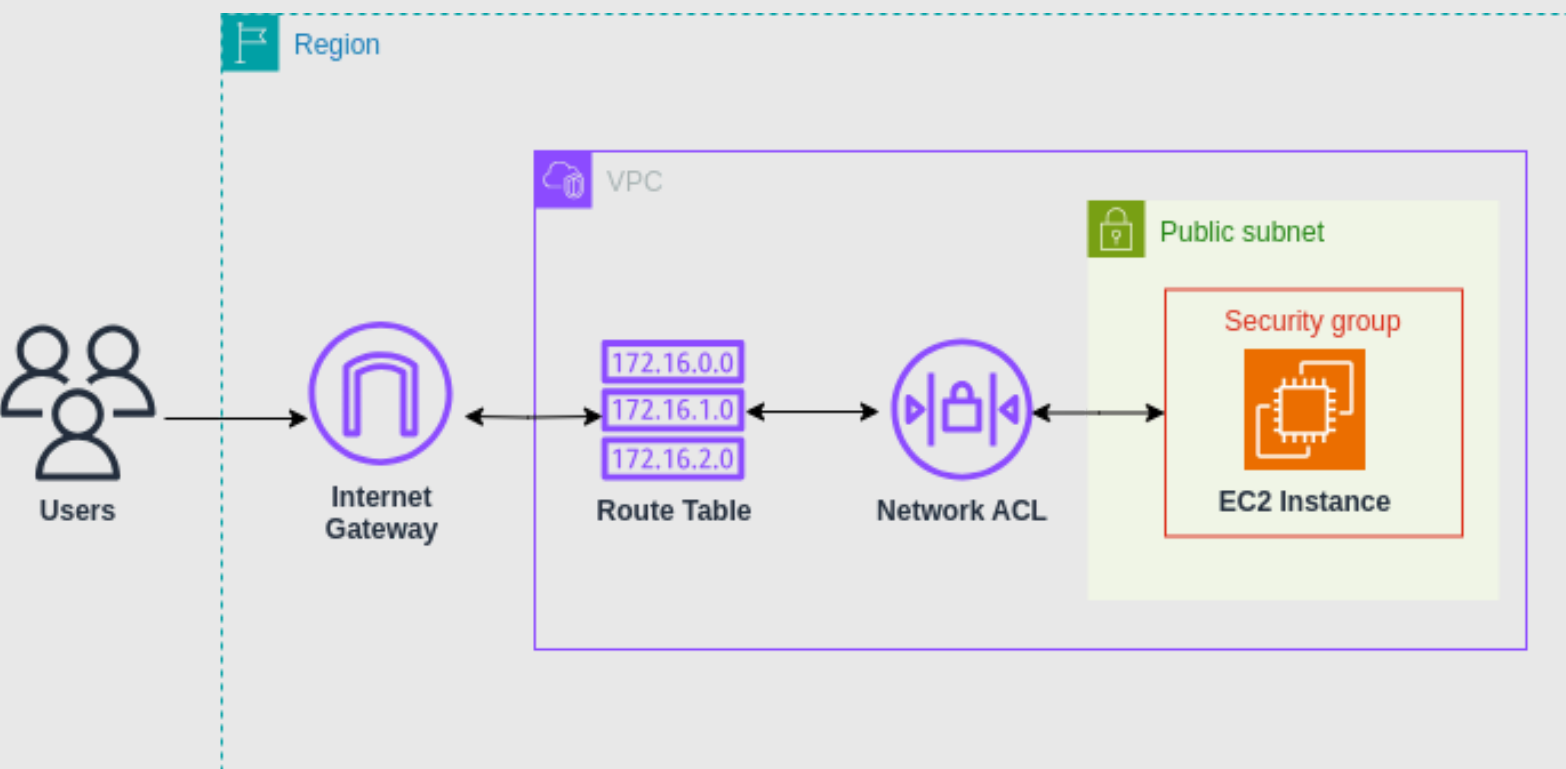
Services

Route Table – Directs network traffic between subnets and external networks.

Security Group – Acts as a virtual firewall for EC2 instances, controlling inbound and outbound traffic.

Network ACL (NACL) – Adds subnet-level security by allowing or denying specific traffic rules.

Architecture Diagram



The architecture has a VPC with subnets, a Route Table for traffic flow, Security Groups for instance level protection, and Network ACLs for subnet-level filtering, ensuring secure and organized communication.

Steps to Implement

1. Create a VPC:

Start by creating your own Virtual Private Cloud (VPC), which is like setting up your own private network inside AWS. You can define the IP address range using CIDR blocks (like 10.0.0.0/16). This step gives you full control over your cloud networking environment like a secure container for all your AWS resources

2. Set Up a Route Table:

Create a route table to manage how traffic moves within the VPC and to/from the internet. Associate it with your subnets and add routes to allow communication (e.g., 0.0.0.0/0 to an internet gateway). This directs network traffic like a GPS.

3. Configure Security Groups:

Launch EC2 instances and attach security groups to control who can access them. Add inbound rules (like allowing SSH or HTTP) and outbound rules based on your needs. This acts as a virtual firewall for individual instances.

4. Create a Network ACL:

Set up a Network Access Control List (NACL) to filter traffic at the subnet level. Define inbound and outbound rules (like allowing port 80 for HTTP). This provides an extra layer of protection beyond the security group.

Story Time

VPC (Your City Walls):

This is like building strong walls around your city. You choose how big the city is by picking the area (IP range). Only invited guests can come inside your secure city!

Route Table (Your City Map):

This is the map that shows all the roads and paths inside your city. It tells travelers where to go to reach houses or leave the city to the outside world.

Security Group (Your House Guards):

These are the guards at each house who check who can come in or go out. They only let in trusted visitors with the right passes like SSH or HTTP.

Network ACL (Your Neighborhood Gatekeeper):

This is the gatekeeper for the whole neighborhood block, making sure everyone entering or leaving follows the rules. It keeps the area extra safe and controlled.

BY
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THANK YOU!

